Three-Link Planar Biomechanical Model

for Human Gait Analysis

Complete Nonlinear Euler–Lagrange Derivation

H-Walker Cable-Driven Walking Rehabilitation Robot

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1 Introduction

This document presents a complete derivation of the nonlinear equations of motion for a three-link planar biomechanical model of the human lower limb during gait. The model represents the **thigh**, **shank**, and **foot** as three rigid links connected by revolute joints at the **hip**, **knee**, and **ankle**.

1.1 Purpose

- Provide a mathematically rigorous model for the H-Walker cable-driven rehabilitation robot.
- Enable model-based control: impedance control, ILC (Iterative Learning Control), and MPC (Model Predictive Control).
- Compute inverse dynamics torques from measured joint trajectories.
- Map cable tension forces to equivalent joint torques.

1.2 Assumptions

1. **Planar (2D) sagittal plane model**: All motion occurs in the sagittal plane. Abduction/adduction and internal/external rotation are neglected.
2. **Rigid body segments**: Each segment (thigh, shank, foot) is a rigid body with constant mass, length, and inertia.
3. **Frictionless joints**: Joint friction is neglected (can be added as viscous damping $\tau_{\text{friction},i} = -b_i\dot{\theta}_i$).
4. **Fixed hip**: During swing phase analysis, the hip is treated as a fixed pivot point. During stance, hip motion can be prescribed.
5. **Point masses at CoM**: The mass of each segment is concentrated at its center of mass, with a rotational inertia about the CoM.

1.3 Notation Summary

Table 1: Symbol definitions used throughout this document.

Symbol	Description	Unit
m_i	Mass of segment i	kg
l_i	Length of segment i	m
l_{c_i}	Distance from proximal joint to CoM of segment i	m
I_i	Moment of inertia of segment i about its CoM	kg·m^2
θ_i	Absolute angle of segment i from downward vertical	rad
$\dot{\theta}_i$	Angular velocity of segment i	rad/s
$\ddot{\theta}_i$	Angular acceleration of segment i	rad/s^2
τ_i	Generalized torque at joint i	N·m
g	Gravitational acceleration (9.81)	m/s^2
s_i	$\sin(\theta_i)$	–
c_i	$\cos(\theta_i)$	–
s_{ij}	$\sin(\theta_i - \theta_j)$	–
c_{ij}	$\cos(\theta_i - \theta_j)$	–
$i = 1$	Thigh (hip $\rightarrow$ knee)	–
$i = 2$	Shank (knee $\rightarrow$ ankle)	–
$i = 3$	Foot (ankle $\rightarrow$ toe)	–

2 Coordinate System and Generalized Coordinates

2.1 Reference Frame

We adopt a right-handed Cartesian coordinate system with:

- Origin at the **hip joint**
- x-axis pointing **forward** (direction of walking)
- y-axis pointing **upward**

2.2 Generalized Coordinates

The system has **3 degrees of freedom**. We use *absolute angles* measured from the downward vertical:

Generalized Coordinates

$$\boldsymbol{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} \tag{1}$$

where each θ_i is measured from the **negative y-axis** (downward vertical), positive in the **counterclockwise** direction (forward flexion).

- $\theta_i = 0$: Segment i hangs straight down (equilibrium)
- $\theta_i > 0$: Segment rotated forward (flexion)
- $\theta_i < 0$: Segment rotated backward (extension)

2.3 Relationship to Clinical Joint Angles

Clinical gait analysis uses *relative* (joint) angles. The conversion is:

$$\phi_{\text{hip}} = \theta_1 \tag{2}$$
$$\phi_{\text{knee}} = \theta_1 - \theta_2 \tag{3}$$
$$\phi_{\text{ankle}} = \theta_2 - \theta_3 + \frac{\pi}{2} \tag{4}$$

Inverse:

$$\theta_1 = \phi_{\text{hip}} \tag{5}$$
$$\theta_2 = \phi_{\text{hip}} - \phi_{\text{knee}} \tag{6}$$
$$\theta_3 = \phi_{\text{hip}} - \phi_{\text{knee}} - \phi_{\text{ankle}} + \frac{\pi}{2} \tag{7}$$

2.4 Model Diagram

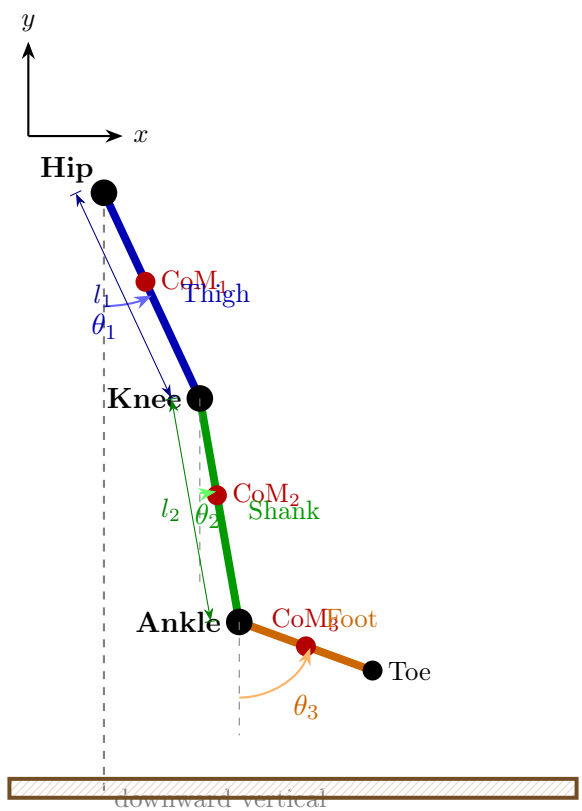

Figure 1: Three-link planar model of the lower limb. Angles θ_1, θ_2, θ_3 are absolute angles measured from the downward vertical (positive = CCW). Red dots indicate centers of mass.

3 Position Kinematics

With the hip at the origin, we derive the position of every joint and center of mass as functions of the generalized coordinates $\boldsymbol{q}$.

3.1 Joint Positions

Joint Position Vectors

Hip (origin):

$$\boldsymbol{p}_{\text{hip}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{8}$$

Knee:

$$\boldsymbol{p}_{\text{knee}} = \begin{bmatrix} l_1 \sin\theta_1 \\ -l_1 \cos\theta_1 \end{bmatrix} \tag{9}$$

Ankle:

$$\boldsymbol{p}_{\text{ankle}} = \begin{bmatrix} l_1 \sin\theta_1 + l_2 \sin\theta_2 \\ -l_1 \cos\theta_1 - l_2 \cos\theta_2 \end{bmatrix} \tag{10}$$

Toe (end of foot):

$$\boldsymbol{p}_{\text{toe}} = \begin{bmatrix} l_1 \sin\theta_1 + l_2 \sin\theta_2 + l_3 \sin\theta_3 \\ -l_1 \cos\theta_1 - l_2 \cos\theta_2 - l_3 \cos\theta_3 \end{bmatrix} \tag{11}$$

3.2 Center of Mass Positions

Each segment's CoM is located at a distance l_{c_i} from its proximal joint, along the segment's axis.

CoM Position Vectors

Thigh CoM:

$$\boldsymbol{p}_{c_1} = \begin{bmatrix} l_{c_1} \sin\theta_1 \\ -l_{c_1} \cos\theta_1 \end{bmatrix} \tag{12}$$

Shank CoM:

$$\boldsymbol{p}_{c_2} = \begin{bmatrix} l_1 \sin\theta_1 + l_{c_2} \sin\theta_2 \\ -l_1 \cos\theta_1 - l_{c_2} \cos\theta_2 \end{bmatrix} \tag{13}$$

Foot CoM:

$$\boldsymbol{p}_{c_3} = \begin{bmatrix} l_1 \sin\theta_1 + l_2 \sin\theta_2 + l_{c_3} \sin\theta_3 \\ -l_1 \cos\theta_1 - l_2 \cos\theta_2 - l_{c_3} \cos\theta_3 \end{bmatrix} \tag{14}$$

4 Velocity Kinematics

The velocity of each CoM is obtained by differentiating the position vectors with respect to time.

4.1 CoM Velocities (Fully Expanded)

Thigh CoM velocity:

$$\boxed{\dot{\boldsymbol{p}}_{c_1} = \begin{bmatrix} l_{c_1} \cos\theta_1 \cdot \dot{\theta}_1 \\ l_{c_1} \sin\theta_1 \cdot \dot{\theta}_1 \end{bmatrix}} \tag{15}$$

Shank CoM velocity:

$$\boxed{\dot{\boldsymbol{p}}_{c_2} = \begin{bmatrix} l_1 \cos\theta_1 \cdot \dot{\theta}_1 + l_{c_2} \cos\theta_2 \cdot \dot{\theta}_2 \\ l_1 \sin\theta_1 \cdot \dot{\theta}_1 + l_{c_2} \sin\theta_2 \cdot \dot{\theta}_2 \end{bmatrix}} \tag{16}$$

Foot CoM velocity:

$$\boxed{\dot{\boldsymbol{p}}_{c_3} = \begin{bmatrix} l_1 \cos\theta_1 \cdot \dot{\theta}_1 + l_2 \cos\theta_2 \cdot \dot{\theta}_2 + l_{c_3} \cos\theta_3 \cdot \dot{\theta}_3 \\ l_1 \sin\theta_1 \cdot \dot{\theta}_1 + l_2 \sin\theta_2 \cdot \dot{\theta}_2 + l_{c_3} \sin\theta_3 \cdot \dot{\theta}_3 \end{bmatrix}} \tag{17}$$

4.2 Squared Speed of Each CoM

For the kinetic energy, we need $v_i^2 = \dot{x}_{c_i}^2 + \dot{y}_{c_i}^2$:

Thigh:

$$v_1^2 = l_{c_1}^2\,\dot{\theta}_1^2 \tag{18}$$

Shank (expanding and using $\cos^2\alpha + \sin^2\alpha = 1$ and $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha-\beta)$):

$$v_2^2 = l_1^2\,\dot{\theta}_1^2 + l_{c_2}^2\,\dot{\theta}_2^2 + 2l_1 l_{c_2}\cos(\theta_1-\theta_2)\,\dot{\theta}_1\dot{\theta}_2 \tag{19}$$

Foot:

$$\boxed{\begin{aligned} v_3^2 = {} & l_1^2\,\dot{\theta}_1^2 + l_2^2\,\dot{\theta}_2^2 + l_{c_3}^2\,\dot{\theta}_3^2 \\ & + 2l_1 l_2\cos(\theta_1-\theta_2)\,\dot{\theta}_1\dot{\theta}_2 \\ & + 2l_1 l_{c_3}\cos(\theta_1-\theta_3)\,\dot{\theta}_1\dot{\theta}_3 \\ & + 2l_2 l_{c_3}\cos(\theta_2-\theta_3)\,\dot{\theta}_2\dot{\theta}_3 \end{aligned}} \tag{20}$$

4.3 Endpoint Jacobian

The Jacobian of the foot endpoint (toe) maps joint velocities to Cartesian velocity:

$$\boldsymbol{J} = \frac{\partial \boldsymbol{p}_{\text{toe}}}{\partial \boldsymbol{q}} = \begin{bmatrix} l_1\cos\theta_1 & l_2\cos\theta_2 & l_3\cos\theta_3 \\ l_1\sin\theta_1 & l_2\sin\theta_2 & l_3\sin\theta_3 \end{bmatrix} \tag{21}$$

This relates: $\dot{\boldsymbol{p}}_{\text{toe}} = \boldsymbol{J}(\boldsymbol{q})\,\dot{\boldsymbol{q}}$ and maps endpoint forces to joint torques: $\boldsymbol{\tau} = \boldsymbol{J}^T\boldsymbol{F}_{\text{ext}}$.

5 Energy Functions

5.1 Kinetic Energy

The total kinetic energy includes translational and rotational contributions:

$$T = \sum_{i=1}^{3}\left(\frac{1}{2}m_i v_i^2 + \frac{1}{2}I_i\dot{\theta}_i^2\right) \tag{22}$$

Substituting the squared speeds from eqs. (18) to (20):

Kinetic Energy — Fully Expanded

$$\begin{aligned} T = {} & \frac{1}{2}\big(m_1 l_{c_1}^2 + m_2 l_1^2 + m_3 l_1^2 + I_1\big)\dot{\theta}_1^2 \\ & + \frac{1}{2}\big(m_2 l_{c_2}^2 + m_3 l_2^2 + I_2\big)\dot{\theta}_2^2 \\ & + \frac{1}{2}\big(m_3 l_{c_3}^2 + I_3\big)\dot{\theta}_3^2 \\ & + \big(m_2 l_1 l_{c_2} + m_3 l_1 l_2\big)\cos(\theta_1-\theta_2)\,\dot{\theta}_1\dot{\theta}_2 \\ & + m_3 l_1 l_{c_3}\cos(\theta_1-\theta_3)\,\dot{\theta}_1\dot{\theta}_3 \\ & + m_3 l_2 l_{c_3}\cos(\theta_2-\theta_3)\,\dot{\theta}_2\dot{\theta}_3 \end{aligned} \tag{23}$$

Note that $T = \frac{1}{2}\dot{\boldsymbol{q}}^T\boldsymbol{M}(\boldsymbol{q})\,\dot{\boldsymbol{q}}$ where $\boldsymbol{M}(\boldsymbol{q})$ is the mass matrix (to be identified in section 7).

5.2 Potential Energy

With the y-axis pointing upward and the hip at the origin:

$$V = \sum_{i=1}^{3} m_i g\, y_{c_i} \tag{24}$$

Substituting the y-coordinates from eqs. (12) to (14):

Potential Energy — Fully Expanded

$$\begin{aligned} V &= -m_1 g\, l_{c_1} \cos\theta_1 \\ &\quad - m_2 g\big(l_1 \cos\theta_1 + l_{c_2} \cos\theta_2\big) \\ &\quad - m_3 g\big(l_1 \cos\theta_1 + l_2 \cos\theta_2 + l_{c_3} \cos\theta_3\big) \\ &= -\big(m_1 l_{c_1} + m_2 l_1 + m_3 l_1\big) g \cos\theta_1 \\ &\quad - \big(m_2 l_{c_2} + m_3 l_2\big) g \cos\theta_2 \\ &\quad - m_3 l_{c_3}\, g \cos\theta_3 \end{aligned} \tag{25}$$

Note: The potential energy reference is at the hip height ($V = 0$ at hip). At $\theta_i = 0$ (hanging down), V reaches its minimum $V_{\min} = -(m_1 l_{c_1} + m_2(l_1 + l_{c_2}) + m_3(l_1 + l_2 + l_{c_3}))g < 0$. Although $V \neq 0$ at equilibrium, $\boldsymbol{G}(\mathbf{0}) = \mathbf{0}$ (no torque needed).

5.3 Lagrangian

$$\mathcal{L}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) - V(\boldsymbol{q}) \tag{26}$$

6 Euler–Lagrange Equations of Motion

The equations of motion are obtained from the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} - \frac{\partial \mathcal{L}}{\partial \theta_i} = \tau_i, \qquad i = 1, 2, 3 \tag{27}$$

This yields the standard manipulator form:

Equations of Motion — Standard Form

$$\boldsymbol{M}(\boldsymbol{q})\, \ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\, \dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{28}$$

where:

- $\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{3\times 3}$: Mass (inertia) matrix — symmetric positive definite
- $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{3\times 3}$: Coriolis/centrifugal matrix
- $\boldsymbol{G}(\boldsymbol{q}) \in \mathbb{R}^{3}$: Gravity vector
- $\boldsymbol{\tau} \in \mathbb{R}^{3}$: Generalized torques $[\tau_1, \tau_2, \tau_3]^T$

6.1 Step-by-Step Derivation for θ_1

To illustrate the derivation process, we work through the first equation in detail.

Step 1: Compute $\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}$ from eq. (23):

$$\begin{aligned}\frac{\partial T}{\partial \dot{\theta}_1} &= \left(m_1 l_{c_1}^2 + m_2 l_1^2 + m_3 l_1^2 + I_1\right)\dot{\theta}_1 \\ &\quad + \left(m_2 l_1 l_{c_2} + m_3 l_1 l_2\right)\cos(\theta_1 - \theta_2)\,\dot{\theta}_2 \\ &\quad + m_3 l_1 l_{c_3}\cos(\theta_1 - \theta_3)\,\dot{\theta}_3 \end{aligned} \tag{29}$$

Step 2: Compute $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}$:

$$\begin{aligned}\frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}_1} &= \underbrace{\left(m_1 l_{c_1}^2 + m_2 l_1^2 + m_3 l_1^2 + I_1\right)}_{\text{constant}}\ddot{\theta}_1 \\ &\quad + \left(m_2 l_1 l_{c_2} + m_3 l_1 l_2\right)\Big[\cos(\theta_1 - \theta_2)\,\ddot{\theta}_2 \\ &\qquad - \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)\dot{\theta}_2\Big] \\ &\quad + m_3 l_1 l_{c_3}\Big[\cos(\theta_1 - \theta_3)\,\ddot{\theta}_3 \\ &\qquad - \sin(\theta_1 - \theta_3)(\dot{\theta}_1 - \dot{\theta}_3)\dot{\theta}_3\Big] \end{aligned} \tag{30}$$

Step 3: Compute $\frac{\partial \mathcal{L}}{\partial \theta_1}$:

$$\begin{aligned}\frac{\partial T}{\partial \theta_1} &= -\left(m_2 l_1 l_{c_2} + m_3 l_1 l_2\right)\sin(\theta_1 - \theta_2)\,\dot{\theta}_1\dot{\theta}_2 \\ &\quad - m_3 l_1 l_{c_3}\sin(\theta_1 - \theta_3)\,\dot{\theta}_1\dot{\theta}_3 \end{aligned} \tag{31}$$

$$\frac{\partial V}{\partial \theta_1} = \left(m_1 l_{c_1} + m_2 l_1 + m_3 l_1\right) g \sin\theta_1 \tag{32}$$

Step 4: Combine: $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} - \frac{\partial \mathcal{L}}{\partial \theta_1} = \tau_1$

After careful cancellation of cross-terms (the $\dot{\theta}_1\dot{\theta}_2$ terms from Steps 2 and 3 cancel exactly), we obtain:

$$\begin{aligned}\tau_1 &= \underbrace{\left(m_1 l_{c_1}^2 + (m_2 + m_3) l_1^2 + I_1\right)}_{M_{11}}\ddot{\theta}_1 \\ &\quad + \underbrace{\left(m_2 l_1 l_{c_2} + m_3 l_1 l_2\right)\cos(\theta_1 - \theta_2)}_{M_{12}}\ddot{\theta}_2 \\ &\quad + \underbrace{m_3 l_1 l_{c_3}\cos(\theta_1 - \theta_3)}_{M_{13}}\ddot{\theta}_3 \\ &\quad \underbrace{-\left(m_2 l_1 l_{c_2} + m_3 l_1 l_2\right)\sin(\theta_1 - \theta_2)\,\dot{\theta}_2^2}_{h_{12}\dot{\theta}_2^2} \\ &\quad \underbrace{-m_3 l_1 l_{c_3}\sin(\theta_1 - \theta_3)\,\dot{\theta}_3^2}_{h_{13}\dot{\theta}_3^2} \\ &\quad + \underbrace{\left(m_1 l_{c_1} + m_2 l_1 + m_3 l_1\right) g \sin\theta_1}_{G_1} \end{aligned} \tag{33}$$

The same procedure for θ_2 and θ_3 yields the remaining two equations (presented in the next subsections).

7 Mass Matrix $\boldsymbol{M}(\boldsymbol{q})$

The mass matrix is identified from the coefficients of $\ddot{\theta}_j$ in each equation. Since $T = \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}\dot{\boldsymbol{q}}$, the elements of $\boldsymbol{M}$ can be read directly from eq. (23).

Mass Matrix — All 9 Elements

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix} \tag{34}$$

$$M_{11} = m_1 l_{c_1}^2 + (m_2 + m_3) l_1^2 + I_1 \tag{35}$$

$$M_{12} = M_{21} = \big(m_2 l_{c_2} + m_3 l_2\big) l_1 \cos(\theta_1 - \theta_2) \tag{36}$$

$$M_{13} = M_{31} = m_3 l_{c_3} l_1 \cos(\theta_1 - \theta_3) \tag{37}$$

$$M_{22} = m_2 l_{c_2}^2 + m_3 l_2^2 + I_2 \tag{38}$$

$$M_{23} = M_{32} = m_3 l_{c_3} l_2 \cos(\theta_2 - \theta_3) \tag{39}$$

$$M_{33} = m_3 l_{c_3}^2 + I_3 \tag{40}$$

Properties of $\boldsymbol{M}(\boldsymbol{q})$

1. **Symmetric**: $M_{ij} = M_{ji}$ (follows from T being a quadratic form in $\dot{\boldsymbol{q}}$).
2. **Positive definite**: For all $\boldsymbol{q}$, $\dot{\boldsymbol{q}}^T \boldsymbol{M}\dot{\boldsymbol{q}} > 0$ when $\dot{\boldsymbol{q}} \neq 0$ (physical: non-zero velocity implies non-zero kinetic energy).
3. **Configuration-dependent**: Only the off-diagonal elements depend on $\boldsymbol{q}$ (through cosine terms). The diagonal elements are *constants*.
4. **Bounded**: $\lambda_{\min}(\boldsymbol{M}) > 0$ for all $\boldsymbol{q}$ (invertibility guaranteed).

Compact Notation

Defining coupling coefficients:

$$\alpha_{12} \triangleq (m_2 l_{c_2} + m_3 l_2) l_1, \quad \alpha_{13} \triangleq m_3 l_{c_3} l_1, \quad \alpha_{23} \triangleq m_3 l_{c_3} l_2 \tag{41}$$

The mass matrix becomes:

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} M_{11}^* & \alpha_{12} c_{12} & \alpha_{13} c_{13} \\ \alpha_{12} c_{12} & M_{22}^* & \alpha_{23} c_{23} \\ \alpha_{13} c_{13} & \alpha_{23} c_{23} & M_{33}^* \end{bmatrix} \tag{42}$$

where M_{ii}^* are the constant diagonal terms and $c_{ij} = \cos(\theta_i - \theta_j)$.

Physical Interpretation

- M_{11}: Total inertia "seen" at the hip. Includes thigh's own inertia plus the inertia of shank and foot (treated as point masses at the knee).
- M_{12}: Coupling between hip and knee. Depends on the relative angle $\theta_1 - \theta_2$; maximum when thigh and shank are aligned ($c_{12} = 1$), zero when perpendicular ($c_{12} = 0$).

- M_{13}: Coupling between hip and ankle. Usually small because the foot mass m_3 is small.
- M_{22}: Total inertia at the knee. Smaller than M_{11} because only shank and foot contribute.
- M_{23}: Coupling between knee and ankle.
- M_{33}: Inertia at the ankle. Smallest element (foot only).

8 Coriolis and Centrifugal Matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$

8.1 Christoffel Symbols

The Coriolis matrix is derived using the Christoffel symbols of the first kind:

$$c_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right) \tag{43}$$

The elements of $\boldsymbol{C}$ are:

$$C_{ij} = \sum_{k=1}^{3} c_{ijk}\,\dot{\theta}_k \tag{44}$$

8.2 Required Partial Derivatives

Since the diagonal elements M_{11}, M_{22}, M_{33} are constants, their partial derivatives vanish. We need:

$$\frac{\partial M_{12}}{\partial \theta_1} = -\alpha_{12}\sin(\theta_1 - \theta_2) \tag{45}$$

$$\frac{\partial M_{12}}{\partial \theta_2} = +\alpha_{12}\sin(\theta_1 - \theta_2) \tag{46}$$

$$\frac{\partial M_{13}}{\partial \theta_1} = -\alpha_{13}\sin(\theta_1 - \theta_3) \tag{47}$$

$$\frac{\partial M_{13}}{\partial \theta_3} = +\alpha_{13}\sin(\theta_1 - \theta_3) \tag{48}$$

$$\frac{\partial M_{23}}{\partial \theta_2} = -\alpha_{23}\sin(\theta_2 - \theta_3) \tag{49}$$

$$\frac{\partial M_{23}}{\partial \theta_3} = +\alpha_{23}\sin(\theta_2 - \theta_3) \tag{50}$$

All other partial derivatives of M_{ij} are zero.

8.3 Coupling Terms

Define the velocity-independent coupling terms:

$$\boxed{\begin{aligned} h_{12} &\triangleq -\alpha_{12}\sin(\theta_1 - \theta_2) = -\big(m_2 l_{c_2} + m_3 l_2\big) l_1 \sin(\theta_1 - \theta_2) \\ h_{13} &\triangleq -\alpha_{13}\sin(\theta_1 - \theta_3) = -m_3 l_{c_3} l_1 \sin(\theta_1 - \theta_3) \\ h_{23} &\triangleq -\alpha_{23}\sin(\theta_2 - \theta_3) = -m_3 l_{c_3} l_2 \sin(\theta_2 - \theta_3) \end{aligned}} \tag{51}$$

8.4 Full Coriolis Matrix

Using Christoffel symbols, the unique Coriolis matrix that guarantees the **skew-symmetry** property $\dot{\boldsymbol{M}} - 2\boldsymbol{C}$ is skew-symmetric:

Coriolis Matrix — All 9 Elements

$$\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & h_{12}\dot{\theta}_2 & h_{13}\dot{\theta}_3 \\ -h_{12}\dot{\theta}_1 & 0 & h_{23}\dot{\theta}_3 \\ -h_{13}\dot{\theta}_1 & -h_{23}\dot{\theta}_2 & 0 \end{bmatrix} \tag{52}$$

Verification: Skew-symmetry of $\dot{\boldsymbol{M}} - 2\boldsymbol{C}$

This property is crucial for passivity-based control design. Define $\boldsymbol{N} = \dot{\boldsymbol{M}} - 2\boldsymbol{C}$. Then for any $\boldsymbol{x} \in \mathbb{R}^3$:

$$\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x} = \boldsymbol{x}^T (\dot{\boldsymbol{M}} - 2\boldsymbol{C}) \boldsymbol{x} = 0 \tag{53}$$

This ensures that the total energy rate depends only on the applied torques: $\dot{E} = \dot{\boldsymbol{q}}^T \boldsymbol{\tau}$ (power balance).

8.5 Coriolis/Centrifugal Force Vector

The vector $\boldsymbol{b} = \boldsymbol{C}\dot{\boldsymbol{q}}$ (which appears directly in the EOM):

Coriolis/Centrifugal Vector $\boldsymbol{b}$

$$b_1 = h_{12}\,\dot{\theta}_2^2 + h_{13}\,\dot{\theta}_3^2 \tag{54}$$

$$b_2 = -h_{12}\,\dot{\theta}_1^2 + h_{23}\,\dot{\theta}_3^2 \tag{55}$$

$$b_3 = -h_{13}\,\dot{\theta}_1^2 - h_{23}\,\dot{\theta}_2^2 \tag{56}$$

Physical Interpretation

- $h_{12}\dot{\theta}_2^2$: **Centrifugal** force on the thigh due to shank rotation. When the shank swings ($\dot{\theta}_2 \neq 0$), it pulls on the thigh through the knee joint.
- $h_{13}\dot{\theta}_3^2$: Centrifugal force on the thigh due to foot rotation (small in practice because m_3 is small).
- $-h_{12}\dot{\theta}_1^2$: Reaction centrifugal force on the shank due to thigh rotation. Note the sign reversal (Newton's 3rd law).
- $h_{23}\dot{\theta}_3^2$: Centrifugal force on the shank due to foot rotation.
- $-h_{13}\dot{\theta}_1^2$, $-h_{23}\dot{\theta}_2^2$: Forces on the foot from thigh and shank rotation.

Note: In this formulation, the diagonal elements of $\boldsymbol{C}$ are zero. The true centrifugal effects appear in the off-diagonal terms when multiplied by $\dot{\boldsymbol{q}}$.

9 Gravity Vector $\boldsymbol{G}(\boldsymbol{q})$

The gravity vector is obtained from $G_i = \frac{\partial V}{\partial \theta_i}$:

Gravity Vector — All 3 Elements

$$G_1 = \big(m_1 l_{c_1} + m_2 l_1 + m_3 l_1\big) g \sin\theta_1 \tag{57}$$

$$G_2 = \big(m_2 l_{c_2} + m_3 l_2\big) g \sin\theta_2 \tag{58}$$

$$G_3 = m_3 l_{c_3}\, g \sin\theta_3 \tag{59}$$

Physical Interpretation

- G_1: Gravitational torque about the hip from *all three segments*. The factor $(m_1 l_{c_1} + m_2 l_1 + m_3 l_1)$ represents the total "gravitational moment arm" — the thigh's own CoM contribution plus the full weight of shank and foot acting at the knee (distance l_1).
- G_2: Gravitational torque about the knee from shank and foot only. The thigh does not contribute because it is proximal to the knee.
- G_3: Gravitational torque about the ankle from the foot only. This is the smallest gravity term.

Equilibrium Check

At $\boldsymbol{q} = \boldsymbol{0}$ (all segments hanging straight down):

$$\boldsymbol{G}(\boldsymbol{0}) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{60}$$

This is correct: no torque is needed to maintain the vertical configuration (it is an equilibrium).

At $\theta_1 = \pi/2$ (thigh horizontal, others hanging):

$$G_1 = \big(m_1 l_{c_1} + (m_2 + m_3) l_1\big) g \tag{61}$$

This is the maximum gravitational load at the hip.

10 Complete Equations of Motion (Fully Expanded)

Combining the mass matrix, Coriolis vector, and gravity vector, we write the three equations of motion in their fully expanded scalar form.

Equation 1 — Hip Joint (θ_1)

$$\begin{aligned}
\tau_1 = {} & \big(m_1 l_{c_1}^2 + (m_2 + m_3) l_1^2 + I_1\big)\ddot{\theta}_1 \\
& + \big(m_2 l_1 l_{c_2} + m_3 l_1 l_2\big) \cos(\theta_1 - \theta_2)\, \ddot{\theta}_2 \\
& + m_3 l_1 l_{c_3} \cos(\theta_1 - \theta_3)\, \ddot{\theta}_3 \\
& - \big(m_2 l_1 l_{c_2} + m_3 l_1 l_2\big) \sin(\theta_1 - \theta_2)\, \dot{\theta}_2^2 \\
& - m_3 l_1 l_{c_3} \sin(\theta_1 - \theta_3)\, \dot{\theta}_3^2 \\
& + \big(m_1 l_{c_1} + m_2 l_1 + m_3 l_1\big) g \sin\theta_1
\end{aligned} \tag{62}$$

Equation 2 — Knee Joint (θ_2)

$$\begin{aligned}\tau_2 = &\big(m_2 l_1 l_{c_2} + m_3 l_1 l_2\big)\cos(\theta_1-\theta_2)\,\ddot{\theta}_1\\ &+\big(m_2 l_{c_2}^2 + m_3 l_2^2 + I_2\big)\ddot{\theta}_2\\ &+ m_3 l_2 l_{c_3}\cos(\theta_2-\theta_3)\,\ddot{\theta}_3\\ &+\big(m_2 l_1 l_{c_2} + m_3 l_1 l_2\big)\sin(\theta_1-\theta_2)\,\dot{\theta}_1^2\\ &- m_3 l_2 l_{c_3}\sin(\theta_2-\theta_3)\,\dot{\theta}_3^2\\ &+\big(m_2 l_{c_2} + m_3 l_2\big) g\sin\theta_2 \end{aligned} \tag{63}$$

Equation 3 — Ankle Joint (θ_3)

$$\begin{aligned}\tau_3 = &\, m_3 l_1 l_{c_3}\cos(\theta_1-\theta_3)\,\ddot{\theta}_1\\ &+ m_3 l_2 l_{c_3}\cos(\theta_2-\theta_3)\,\ddot{\theta}_2\\ &+\big(m_3 l_{c_3}^2 + I_3\big)\ddot{\theta}_3\\ &+ m_3 l_1 l_{c_3}\sin(\theta_1-\theta_3)\,\dot{\theta}_1^2\\ &+ m_3 l_2 l_{c_3}\sin(\theta_2-\theta_3)\,\dot{\theta}_2^2\\ &+ m_3 l_{c_3}\, g\sin\theta_3 \end{aligned} \tag{64}$$

Structure Analysis

Each equation has the form:

$$\tau_i = \underbrace{\sum_{j=1}^{3} M_{ij}\ddot{\theta}_j}_{\text{inertial}} + \underbrace{b_i(\boldsymbol{q},\dot{\boldsymbol{q}})}_{\text{Coriolis/centrifugal}} + \underbrace{G_i(\boldsymbol{q})}_{\text{gravity}} \tag{65}$$

Key observations:

1. The **inertial terms** (involving $\ddot{\theta}$) are coupled through the off-diagonal mass matrix elements. When the shank accelerates, it creates a torque at the hip (and vice versa).
2. The **Coriolis/centrifugal terms** are proportional to $\dot{\theta}^2$. They are the nonlinear velocity-dependent forces that make the system challenging to control.
3. The **gravity terms** depend only on $\sin\theta_i$ (pendulum-like). They represent the dominant load during stance.
4. The equations are **nonlinear** due to $\sin(\theta_i-\theta_j)$, $\cos(\theta_i-\theta_j)$, and $\dot{\theta}^2$ terms.

11 Forward and Inverse Dynamics

11.1 Forward Dynamics

Given applied torques $\boldsymbol{\tau}$, compute accelerations:

$$\boxed{\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\big[\boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{G}(\boldsymbol{q})\big]} \tag{66}$$

In practice, we solve $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{\tau} - \boldsymbol{b} - \boldsymbol{G}$ using LU decomposition (not explicit matrix inversion).

11.2 Inverse Dynamics

Given desired trajectory $(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t), \ddot{\boldsymbol{q}}(t))$, compute required torques:

$$\boxed{\boldsymbol{\tau} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q})} \tag{67}$$

This is used for:

- Computing feedforward torques in model-based control (ILC, MPC)
- Analyzing joint loading during gait from motion capture data
- Designing the H-Walker cable force profile

11.3 State-Space Form for Simulation

Define the state vector:

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \in \mathbb{R}^6 \tag{68}$$

The state-space ODE is:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\boldsymbol{q}} \\ \boldsymbol{M}^{-1}(\boldsymbol{q})\big[\boldsymbol{\tau} - \boldsymbol{b}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \boldsymbol{G}(\boldsymbol{q})\big] \end{bmatrix} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\tau}) \tag{69}$$

This is integrated numerically using RK45 (Dormand–Prince) with adaptive step size.

11.4 Energy Conservation

For the conservative case ($\boldsymbol{\tau} = \boldsymbol{0}$), the total mechanical energy:

$$E = T + V = \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}} + V(\boldsymbol{q}) = \text{const} \tag{70}$$

This serves as a **verification criterion**: simulating with $\boldsymbol{\tau} = \boldsymbol{0}$ should yield $\frac{dE}{dt} = 0$ (up to numerical integration error).

When external torques are applied, the power balance is:

$$\frac{dE}{dt} = \dot{\boldsymbol{q}}^T \boldsymbol{\tau} \tag{71}$$

12 External Force Application — General Framework

This section presents a **general framework** for computing joint torques and kinematic effects when an external force is applied at *any point* on *any link* of the 3-link model. The H-Walker cable force (section 13) is a specific case of this framework.

12.1 Principle of Virtual Work

An external force $\boldsymbol{F} = [F_x, F_y]^T$ applied at point $\boldsymbol{p}$ on the mechanism produces generalized forces (joint torques) via:

$$\boxed{\tau_i^{\text{ext}} = \boldsymbol{F}^T \cdot \frac{\partial \boldsymbol{p}}{\partial \theta_i}, \qquad i = 1, 2, 3} \tag{72}$$

In matrix form, using the **point Jacobian** $\boldsymbol{J}_p = \partial \boldsymbol{p} / \partial \boldsymbol{q}$:

$$\boldsymbol{\tau}^{\text{ext}} = \boldsymbol{J}_p^T \boldsymbol{F} \tag{73}$$

The EOM with external force becomes:

$$\boxed{\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{J}_p^T \boldsymbol{F}_{\text{ext}}} \tag{74}$$

12.2 Case 1: Force Applied on the Thigh

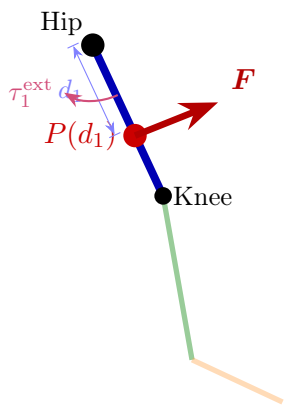

Figure 2: External force on the thigh at distance d_1 from the hip. Only the hip joint torque τ_1 is generated.

A force applied at distance d_1 from the hip along the thigh:

$$\boldsymbol{p}(d_1) = \begin{bmatrix} d_1 \sin\theta_1 \\ -d_1 \cos\theta_1 \end{bmatrix} \tag{75}$$

Point Jacobian:

$$\boldsymbol{J}_p = \begin{bmatrix} d_1 \cos\theta_1 & 0 & 0 \\ d_1 \sin\theta_1 & 0 & 0 \end{bmatrix} \tag{76}$$

Joint Torques from Force on Thigh

$$\tau_1^{\text{ext}} = d_1(F_x \cos\theta_1 + F_y \sin\theta_1) = d_1 \|\boldsymbol{F}\| \sin\left(\alpha - \theta_1 + \tfrac{\pi}{2}\right) \tag{77}$$

$$\tau_2^{\text{ext}} = 0 \tag{78}$$

$$\tau_3^{\text{ext}} = 0 \tag{79}$$

where $\alpha = \text{atan2}(F_y, F_x)$ is the force direction.
Key insight: A force on the thigh affects *only the hip joint*. The knee and ankle are unaffected because the thigh is proximal to both.

2D Cross Product (Moment Arm) Interpretation

The torque about the hip can also be written as a 2D cross product:

$$\tau_1^{\text{ext}} = \boldsymbol{r} \times \boldsymbol{F} = r_x F_y - r_y F_x = d_1 \sin\theta_1 \cdot F_y - (-d_1 \cos\theta_1) \cdot F_x \tag{80}$$

where $\boldsymbol{r} = \boldsymbol{p} - \boldsymbol{p}_{\text{hip}}$. The effective moment arm is $d_\perp = d_1 \sin\phi$, where ϕ is the angle between the force direction and the thigh axis.

12.3 Case 2: Force Applied on the Shank

A force at distance d_2 from the knee along the shank:

$$\boldsymbol{p}(d_2) = \begin{bmatrix} l_1 \sin\theta_1 + d_2 \sin\theta_2 \\ -l_1 \cos\theta_1 - d_2 \cos\theta_2 \end{bmatrix} \tag{81}$$

Point Jacobian:

$$\boldsymbol{J}_p = \begin{bmatrix} l_1 \cos\theta_1 & d_2 \cos\theta_2 & 0 \\ l_1 \sin\theta_1 & d_2 \sin\theta_2 & 0 \end{bmatrix} \tag{82}$$

Joint Torques from Force on Shank

$$\tau_1^{\text{ext}} = l_1(F_x \cos\theta_1 + F_y \sin\theta_1) \tag{83}$$
$$\tau_2^{\text{ext}} = d_2(F_x \cos\theta_2 + F_y \sin\theta_2) \tag{84}$$
$$\tau_3^{\text{ext}} = 0 \tag{85}$$

Key insight: A force on the shank creates torques at *both hip and knee*, but *not the ankle*. The hip torque depends on the *full thigh length* l_1 (not the point of application), because the entire thigh acts as a rigid lever arm.

Moment Arm Analysis

$$\begin{aligned} \text{Moment arm about hip:} \quad & d_{\perp,\text{hip}} = \left\|\boldsymbol{p}(d_2) - \boldsymbol{p}_{\text{hip}}\right\| \sin\phi_{\text{hip}} \\ \text{Moment arm about knee:} \quad & d_{\perp,\text{knee}} = \left\|\boldsymbol{p}(d_2) - \boldsymbol{p}_{\text{knee}}\right\| \sin\phi_{\text{knee}} = d_2 \sin\phi_{\text{knee}} \end{aligned} \tag{86}$$

where ϕ is the angle between the force and the position vector from the joint.

Torque Ratio

The ratio of hip torque to knee torque:

$$\frac{\tau_1^{\text{ext}}}{\tau_2^{\text{ext}}} = \frac{l_1(F_x \cos\theta_1 + F_y \sin\theta_1)}{d_2(F_x \cos\theta_2 + F_y \sin\theta_2)} \tag{87}$$

When $\theta_1 \approx \theta_2$ (thigh and shank nearly aligned): $\tau_1/\tau_2 \approx l_1/d_2$, which is purely geometric.

12.4 Case 3: Force Applied on the Foot

A force at distance d_3 from the ankle along the foot:

$$\boldsymbol{p}(d_3) = \begin{bmatrix} l_1 \sin\theta_1 + l_2 \sin\theta_2 + d_3 \sin\theta_3 \\ -l_1 \cos\theta_1 - l_2 \cos\theta_2 - d_3 \cos\theta_3 \end{bmatrix} \tag{88}$$

Point Jacobian:

$$\boldsymbol{J}_p = \begin{bmatrix} l_1 \cos\theta_1 & l_2 \cos\theta_2 & d_3 \cos\theta_3 \\ l_1 \sin\theta_1 & l_2 \sin\theta_2 & d_3 \sin\theta_3 \end{bmatrix} \tag{89}$$

Joint Torques from Force on Foot — All 3 Joints Affected

$$\tau_1^{\text{ext}} = l_1(F_x \cos\theta_1 + F_y \sin\theta_1) \tag{90}$$
$$\tau_2^{\text{ext}} = l_2(F_x \cos\theta_2 + F_y \sin\theta_2) \tag{91}$$
$$\tau_3^{\text{ext}} = d_3(F_x \cos\theta_3 + F_y \sin\theta_3) \tag{92}$$

Key insight: A force on the foot creates torques at *all three joints*. The hip and knee torques depend on the *full segment lengths* l_1, l_2 respectively, independent of where on the foot the force is applied. Only the ankle torque depends on the application point d_3.

Special Case: Ground Reaction Force (GRF)

During stance, the GRF acts at the foot-ground contact point (approximately at $d_3 \approx l_3/2$ for mid-foot contact). A typical GRF of $\boldsymbol{F}_{\text{GRF}} = [F_x^{\text{GRF}}, F_y^{\text{GRF}}]^T$ where $F_y^{\text{GRF}} \approx M_{\text{body}} g$ produces:

$$\tau_1^{\text{GRF}} = l_1(F_x^{\text{GRF}} \cos\theta_1 + F_y^{\text{GRF}} \sin\theta_1) \tag{93}$$
$$\tau_2^{\text{GRF}} = l_2(F_x^{\text{GRF}} \cos\theta_2 + F_y^{\text{GRF}} \sin\theta_2) \tag{94}$$
$$\tau_3^{\text{GRF}} = d_3(F_x^{\text{GRF}} \cos\theta_3 + F_y^{\text{GRF}} \sin\theta_3) \tag{95}$$

12.5 General Case: Force at Arbitrary Point with Offset

In practice (e.g., H-Walker cable), forces may be applied at a point that is *offset* from the link axis. For a force applied at:

$$\boldsymbol{p}_{\text{offset}} = \boldsymbol{p}_{\text{joint}} + d_{\parallel} \begin{bmatrix} \sin\theta_k \\ -\cos\theta_k \end{bmatrix} + d_{\perp} \begin{bmatrix} -\cos\theta_k \\ -\sin\theta_k \end{bmatrix} \tag{96}$$

where $d_{\parallel}$ is the distance along the link and $d_{\perp}$ is the perpendicular offset (e.g., posterior side), the Jacobian column for joint k gains an additional term from the offset:

$$\frac{\partial \boldsymbol{p}_{\text{offset}}}{\partial \theta_k} = \begin{bmatrix} d_{\parallel} \cos\theta_k + d_{\perp} \sin\theta_k \\ d_{\parallel} \sin\theta_k - d_{\perp} \cos\theta_k \end{bmatrix} \tag{97}$$

This is exactly the formulation used for the H-Walker cable attachment in section 13.

12.6 Summary: Force Application Rules

Table 2: Which joints are affected by force application at each link.

Force on	τ_1 (Hip)	τ_2 (Knee)	τ_3 (Ankle)	**Jacobian columns**
Thigh (at d_1)	✓	×	×	$[d_1, 0, 0]$
Shank (at d_2)	✓	✓	×	$[l_1, d_2, 0]$
Foot (at d_3)	✓	✓	✓	$[l_1, l_2, d_3]$

Fundamental Rule

A force applied at any point on link k produces torques at joint k and **all joints proximal to it** (joints 1 through k). Joints **distal** to the force application point (joints $k+1$ through 3) are **unaffected**.

The torque at a proximal joint $j < k$ depends on the **full length** l_j of the intervening link, not on where the force is applied. Only the torque at joint k itself depends on the distance d_k from the proximal joint to the force application point.

12.7 Kinematic Response to Applied Force

Given an applied force $\boldsymbol{F}$ at point $\boldsymbol{p}$, the resulting **joint accelerations** (assuming no muscle torque) are:

$$\boxed{\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[\boldsymbol{J}_p^T\boldsymbol{F} - \boldsymbol{C}\dot{\boldsymbol{q}} - \boldsymbol{G}\right]} \tag{98}$$

The **acceleration of the force application point** itself is:

$$\ddot{\boldsymbol{p}} = \boldsymbol{J}_p\ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}_p\dot{\boldsymbol{q}} \tag{99}$$

Operational Space Dynamics

Projecting the dynamics into Cartesian space at the force application point:

$$\boldsymbol{\Lambda}_p(\boldsymbol{q})\,\ddot{\boldsymbol{p}} + \boldsymbol{\mu}_p(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{p}_p(\boldsymbol{q}) = \boldsymbol{F} \tag{100}$$

where $\boldsymbol{\Lambda}_p = (\boldsymbol{J}_p\boldsymbol{M}^{-1}\boldsymbol{J}_p^T)^{-1}$ is the **operational space inertia** (the apparent mass felt at point $\boldsymbol{p}$).

This tells us: if you push on the shank with force $\boldsymbol{F}$, the acceleration you feel is governed by $\boldsymbol{\Lambda}_p$, which depends on the *entire* chain's inertia projected through the Jacobian.

Numerical Example

For a 70 kg, 1.75 m subject in mid-swing ($\theta_1 = 20$, $\theta_2 = 45$, $\theta_3 = 30$), applying $\boldsymbol{F} = [0, -50]^T$ N (50 N downward) at the mid-shank ($d_2 = l_2/2 = 0.215$ m):

$$\begin{aligned}
\tau_1^{\text{ext}} &= 0.4287 \times (0 \cdot \cos 20 + (-50) \cdot \sin 20) = 0.4287 \times (-17.10) = -7.33\ \text{N m} \\
\tau_2^{\text{ext}} &= 0.2153 \times (0 \cdot \cos 45 + (-50) \cdot \sin 45) = 0.2153 \times (-35.36) = -7.61\ \text{N m} \\
\tau_3^{\text{ext}} &= 0
\end{aligned}$$

The 50 N force at mid-shank creates approximately equal torques at hip (-7.33 Nm) and knee (-7.61 Nm), both tending toward extension.

12.8 Multiple Simultaneous Forces

When multiple forces are applied simultaneously (e.g., cable force + GRF + assistive device), the generalized forces simply **superpose**:

$$\boldsymbol{\tau}_{\text{total}}^{\text{ext}} = \sum_k \boldsymbol{J}_{p_k}^T\boldsymbol{F}_k \tag{101}$$

For the H-Walker during stance phase:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = \boldsymbol{\tau}_{\text{muscle}} + \underbrace{\boldsymbol{J}_{\text{cable}}^T\boldsymbol{F}_{\text{cable}}}_{\text{H-Walker assist}} + \underbrace{\boldsymbol{J}_{\text{GRF}}^T\boldsymbol{F}_{\text{GRF}}}_{\text{ground reaction}} \tag{102}$$

13 H-Walker Cable Force Mapping

13.1 Cable Geometry

The H-Walker uses cables from motor spools to assist knee flexion during the swing phase. The cable attaches on the **posterior side of the shank**, below the knee, and routes forward to a **pulley on the walker frame** in front of the hip.

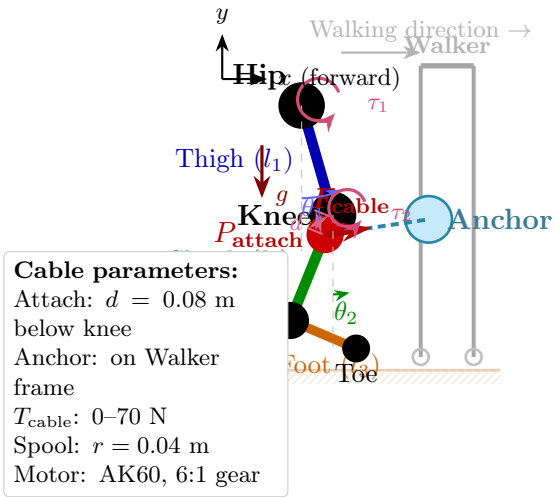

Figure 3: H-Walker cable routing geometry in the sagittal plane. The cable attaches at P_{attach} on the posterior side of the shank (distance d below the knee), and routes forward to a fixed pulley on the walker frame (in front of the person). The cable force $\boldsymbol{F}_{\text{cable}}$ acts along the cable direction, producing torques τ_1 (hip) and τ_2 (knee). Configuration shown: mid-swing phase ($\theta_1 \approx 25$, $\theta_2 \approx 12$).

13.2 Cable Attachment Point

The attachment point in the world frame:

$$\boldsymbol{p}_{\text{attach}} = \boldsymbol{p}_{\text{knee}} + d_{\parallel} \begin{bmatrix} \sin\theta_2 \\ -\cos\theta_2 \end{bmatrix} + d_{\perp} \begin{bmatrix} -\cos\theta_2 \\ -\sin\theta_2 \end{bmatrix} \tag{103}$$

where $d_{\parallel}$ is the distance along the shank and $d_{\perp}$ is the posterior offset.

Expanded:

$$\boldsymbol{p}_{\text{attach}} = \begin{bmatrix} l_1 \sin\theta_1 + d_{\parallel} \sin\theta_2 - d_{\perp} \cos\theta_2 \\ -l_1 \cos\theta_1 - d_{\parallel} \cos\theta_2 - d_{\perp} \sin\theta_2 \end{bmatrix} \tag{104}$$

13.3 Cable Direction and Force Vector

The unit direction from attachment to pulley:

$$\hat{\boldsymbol{u}} = \frac{\boldsymbol{p}_{\text{pulley}} - \boldsymbol{p}_{\text{attach}}}{\|\boldsymbol{p}_{\text{pulley}} - \boldsymbol{p}_{\text{attach}}\|} \tag{105}$$

The cable force vector (tension is always positive, cable can only pull):

$$\boldsymbol{F}_{\text{cable}} = T_{\text{cable}} \cdot \hat{\boldsymbol{u}}, \quad T_{\text{cable}} \geq 0 \tag{106}$$

13.4 Jacobian Mapping: Cable Force to Joint Torques

Using the principle of virtual work, the generalized forces from the cable are:

$$Q_i = \boldsymbol{F}_{\text{cable}} \cdot \frac{\partial \boldsymbol{p}_{\text{attach}}}{\partial \theta_i} \tag{107}$$

Define the cable Jacobian:

$$\boldsymbol{J}_{\text{cable}} = \frac{\partial \boldsymbol{p}_{\text{attach}}}{\partial \boldsymbol{q}} = \begin{bmatrix} l_1 \cos\theta_1 & d_{\parallel} \cos\theta_2 + d_{\perp} \sin\theta_2 & 0 \\ l_1 \sin\theta_1 & d_{\parallel} \sin\theta_2 - d_{\perp} \cos\theta_2 & 0 \end{bmatrix} \tag{108}$$

Note that the third column is zero because the cable attaches above the ankle (i.e., $\boldsymbol{p}_{\text{attach}}$ does not depend on θ_3).

Cable-to-Joint Torque Mapping

$$\boldsymbol{\tau}_{\text{cable}} = \boldsymbol{J}_{\text{cable}}^T \boldsymbol{F}_{\text{cable}} = T_{\text{cable}} \cdot \boldsymbol{J}_{\text{cable}}^T \hat{\boldsymbol{u}} \tag{109}$$

Expanded:

$$\tau_{\text{cable},1} = T_{\text{cable}} \big(l_1 \cos\theta_1 \, \hat{u}_x + l_1 \sin\theta_1 \, \hat{u}_y \big) \tag{110}$$

$$\tau_{\text{cable},2} = T_{\text{cable}} \big((d_{\parallel} \cos\theta_2 + d_{\perp} \sin\theta_2) \hat{u}_x + (d_{\parallel} \sin\theta_2 - d_{\perp} \cos\theta_2) \hat{u}_y \big) \tag{111}$$

$$\tau_{\text{cable},3} = 0 \tag{112}$$

13.5 Motor–Cable–Force Chain

The H-Walker uses AK60 motors. The conversion chain is:

$$\underbrace{I_{\text{motor}}}_{[\text{A}]} \xrightarrow{K_t} \underbrace{\tau_{\text{motor}}}_{[\text{Nm}]} \xrightarrow{N_{\text{gear}} \cdot \eta} \underbrace{\tau_{\text{spool}}}_{[\text{Nm}]} \xrightarrow{1/r_{\text{spool}}} \underbrace{F_{\text{cable}}}_{[\text{N}]} \tag{113}$$

$$\boxed{F_{\text{cable}} = \frac{I_{\text{motor}} \cdot K_t \cdot N_{\text{gear}} \cdot \eta}{r_{\text{spool}}}} \tag{114}$$

With H-Walker parameters:

$$F_{\text{cable}} = \frac{I \times 0.078 \times 6.0 \times 0.869}{0.04} = I \times 10.17 \ [\text{N/A}] \tag{115}$$

14 Normal Gait Reference Trajectories

14.1 Fourier Series Representation

Normal gait joint angles are periodic and well-represented by a Fourier series (Winter, 2009):

$$\theta(t) = a_0 + \sum_{n=1}^{N} \left[a_n \cos\left(\frac{2\pi n t}{T}\right) + b_n \sin\left(\frac{2\pi n t}{T}\right) \right] \tag{116}$$

where T is the gait cycle duration and $N = 6$ harmonics provide excellent accuracy.

14.2 Gait Cycle Phases

Table 3: Normal gait cycle phases (heel strike = 0%).

GCP [%]	Phase	Sub-phase	Key Event
0	Stance	Loading Response	Heel Strike
0–10	Stance	Loading Response	Weight acceptance
10–30	Stance	Mid Stance	Single limb support
30–50	Stance	Terminal Stance	Heel rise
50–60	Stance	Pre-Swing	Toe Off ($\approx 60\%$)
60–73	Swing	Initial Swing	Foot clearance
73–87	Swing	Mid Swing	Limb advancement
87–100	Swing	Terminal Swing	Deceleration

14.3 Normal Joint Angle Profiles

From Winter (2009), natural cadence ($\approx$ 108 steps/min):

Table 4: Key joint angle values during normal gait (degrees).

Event	Hip Flex/Ext	Knee Flex	Ankle Dorsi/Plantar
Heel Strike (0%)	$+25°$	$\approx 5°$	$\approx 0°$
Loading Response (8%)	$+20°$	$\approx 15°$	$-5°$
Mid Stance (30%)	$0°$	$\approx 5°$	$+10°$
Terminal Stance (50%)	$-10°$	$\approx 5°$	$+15°$
Toe Off (60%)	$-5°$	$\approx 40°$	$-20°$
Mid Swing (73%)	$+15°$	$\approx 65°$	$+5°$
Terminal Swing (95%)	$+30°$	$\approx 5°$	$\approx 0°$

14.4 Angular Velocity (Analytical Derivative)

The derivative of the Fourier series gives the angular velocity:

$$\dot{\theta}(t) = \sum_{n=1}^{N} \frac{2\pi n}{T} \left[-a_n \sin\left(\frac{2\pi n t}{T}\right) + b_n \cos\left(\frac{2\pi n t}{T}\right) \right] \tag{117}$$

This analytical derivative avoids numerical differentiation noise, which is critical for accurate inverse dynamics computation.

15 Anthropometric Parameters

15.1 Winter (2009) Body Segment Parameter Data

Segment properties are expressed as ratios of total body mass (M) and height (H):

Table 5: Body segment parameters from Winter (2009) and Dempster (1955).

Segment	Mass/M	Length/H	CoM/Length	ρ/Length
Thigh	0.100	0.245	0.433	0.323
Shank	0.0465	0.246	0.433	0.302
Foot	0.0145	0.152	0.500	0.475

where ρ/Length is the radius of gyration about CoM as a fraction of segment length. The moment of inertia is:

$$I_i = m_i \cdot (\rho_i \cdot l_i)^2 = m_i \cdot \left(\frac{\rho_i}{l_i} \cdot l_i\right)^2 \tag{118}$$

15.2 Numerical Values (70 kg, 1.75 m Adult)

Table 6: Computed segment parameters for a 70 kg, 1.75 m subject.

Segment	m_i [kg]	l_i [m]	l_{c_i} [m]	I_i [kg·m^2]	ρ_i [m]
Thigh ($i = 1$)	7.000	0.4288	0.1857	0.1342	0.1385
Shank ($i = 2$)	3.255	0.4305	0.1864	0.0550	0.1300
Foot ($i = 3$)	1.015	0.2660	0.1330	0.0162	0.1264

15.3 Computed Model Coefficients

Substituting the numerical values into the model equations:

Table 7: Numerical coefficients in the equations of motion.

Coefficient	**Value**	**Expression**
Mass matrix diagonal (constants):		
M_{11}^*	1.160 kg·m^2	$m_1 l_{c_1}^2 + (m_2 + m_3) l_1^2 + I_1$
M_{22}^*	0.356 kg·m^2	$m_2 l_{c_2}^2 + m_3 l_2^2 + I_2$
M_{33}^*	0.034 kg·m^2	$m_3 l_{c_3}^2 + I_3$
Coupling coefficients:		
α_{12}	0.448 kg·m^2	$(m_2 l_{c_2} + m_3 l_2) l_1$
α_{13}	0.058 kg·m^2	$m_3 l_{c_3} l_1$
α_{23}	0.058 kg·m^2	$m_3 l_{c_3} l_2$
Gravity coefficients:		
γ_1	30.71 N·m	$(m_1 l_{c_1} + m_2 l_1 + m_3 l_1) g$
γ_2	10.24 N·m	$(m_2 l_{c_2} + m_3 l_2) g$
γ_3	1.32 N·m	$m_3 l_{c_3} g$

15.4 Equations with Numerical Coefficients

For the 70 kg, 1.75 m subject:

EOM with Numerical Values

$$\begin{aligned}\tau_1 = {} & 1.160\,\ddot{\theta}_1 + 0.448\cos(\theta_1 - \theta_2)\,\ddot{\theta}_2 + 0.058\cos(\theta_1 - \theta_3)\,\ddot{\theta}_3 \\ & - 0.448\sin(\theta_1 - \theta_2)\,\dot{\theta}_2^2 - 0.058\sin(\theta_1 - \theta_3)\,\dot{\theta}_3^2 + 30.71\sin\theta_1\end{aligned} \tag{119}$$

$$\begin{aligned}\tau_2 = {} & 0.448\cos(\theta_1 - \theta_2)\,\ddot{\theta}_1 + 0.356\,\ddot{\theta}_2 + 0.058\cos(\theta_2 - \theta_3)\,\ddot{\theta}_3 \\ & + 0.448\sin(\theta_1 - \theta_2)\,\dot{\theta}_1^2 - 0.058\sin(\theta_2 - \theta_3)\,\dot{\theta}_3^2 + 10.24\sin\theta_2\end{aligned} \tag{120}$$

$$\begin{aligned}\tau_3 = {} & 0.058\cos(\theta_1 - \theta_3)\,\ddot{\theta}_1 + 0.058\cos(\theta_2 - \theta_3)\,\ddot{\theta}_2 + 0.034\,\ddot{\theta}_3 \\ & + 0.058\sin(\theta_1 - \theta_3)\,\dot{\theta}_1^2 + 0.058\sin(\theta_2 - \theta_3)\,\dot{\theta}_2^2 + 1.32\sin\theta_3\end{aligned} \tag{121}$$

16 Linearization for Control Design

For MPC and linear control, we linearize about a reference trajectory $(\boldsymbol{q}_0, \dot{\boldsymbol{q}}_0, \boldsymbol{\tau}_0)$:

$$\boldsymbol{M}_0\,\delta\ddot{\boldsymbol{q}} + \boldsymbol{C}_0\,\delta\dot{\boldsymbol{q}} + \boldsymbol{K}_0\,\delta\boldsymbol{q} = \delta\boldsymbol{\tau} \tag{122}$$

where:

$$\boldsymbol{M}_0 = \boldsymbol{M}(\boldsymbol{q}_0) \tag{123}$$

$$\boldsymbol{C}_0 = \boldsymbol{C}(\boldsymbol{q}_0, \dot{\boldsymbol{q}}_0) \tag{124}$$

$$\boldsymbol{K}_0 = \left.\frac{\partial(\boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G})}{\partial \boldsymbol{q}}\right|_{\boldsymbol{q}_0, \dot{\boldsymbol{q}}_0} \tag{125}$$

The stiffness matrix $\boldsymbol{K}_0$ includes both the gravitational stiffness $(\partial G_i/\partial q_j)$ and the velocity-dependent stiffness from the Coriolis terms.

For small perturbations about the hanging equilibrium ($\boldsymbol{q}_0 = \boldsymbol{0}$, $\dot{\boldsymbol{q}}_0 = \boldsymbol{0}$):

$$K_{ij} = \left.\frac{\partial G_i}{\partial \theta_j}\right|_{\boldsymbol{q}=\boldsymbol{0}} = \gamma_i\,\delta_{ij} \tag{126}$$

which is diagonal — the linearized system decouples when hanging straight down.

17 Model-Based Control Application for H-Walker

This section connects the biomechanical model to the actual H-Walker control problem: *given a fixed cable attachment point, how does the cable force affect joint torques, angular accelerations, and gait parameters (stride length, foot clearance) at each instant?*

17.1 Instantaneous Effect of Cable Force on Gait

At each time instant during gait (characterized by GCP%), the cable force T_{cable} produces specific joint torques that depend on the *current leg configuration* $\boldsymbol{q}(t)$.

Step 1: Cable Force → Joint Torques

From section 13, at configuration $\boldsymbol{q}$:

$$\boldsymbol{\tau}_{\text{cable}}(\boldsymbol{q}, T) = T \cdot \boldsymbol{J}_{\text{cable}}^T(\boldsymbol{q})\,\hat{\boldsymbol{u}}(\boldsymbol{q}) \tag{127}$$

The key insight: **the same cable tension T produces different torques at different gait phases**, because $\boldsymbol{q}$ changes throughout the gait cycle, which changes both the cable direction $\hat{\boldsymbol{u}}$ and the moment arms.

Table 8: Cable force effect across the gait cycle (50 N tension, 70 kg subject).

Phase	**GCP**	θ_1	θ_2	τ_{hip}	τ_{knee}
Pre-swing	60%	−6	17	−4.2 Nm	−0.7 Nm
Initial swing	65%	3	29	−4.2 Nm	−0.7 Nm
Mid-swing	73%	17	46	−1.7 Nm	−0.3 Nm
Terminal swing	90%	23	9	−1.5 Nm	−2.0 Nm

Observation: The cable is most effective for knee flexion assistance during **initial swing** (60–65% GCP) and most effective for knee extension control during **terminal swing** (85–95% GCP).

Step 2: Joint Torques → Angular Accelerations

The cable-induced torques produce angular accelerations via forward dynamics:

$$\ddot{\boldsymbol{q}}_{\text{cable}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\,\boldsymbol{\tau}_{\text{cable}} \tag{128}$$

Note: Due to the mass matrix coupling, the cable force on the shank affects *all three joint accelerations*, not just hip and knee:

$$\begin{bmatrix} \ddot{\theta}_{1,\text{cable}} \\ \ddot{\theta}_{2,\text{cable}} \\ \ddot{\theta}_{3,\text{cable}} \end{bmatrix} = \boldsymbol{M}^{-1} \begin{bmatrix} \tau_{\text{cable},1} \\ \tau_{\text{cable},2} \\ 0 \end{bmatrix} \neq \begin{bmatrix} \tau_{\text{cable},1}/M_{11} \\ \tau_{\text{cable},2}/M_{22} \\ 0 \end{bmatrix} \tag{129}$$

The right side is **wrong** — you cannot divide each torque by its diagonal mass element because $\boldsymbol{M}$ is coupled. The off-diagonal elements M_{31}, M_{32} transmit the effect to the ankle even though $\tau_{\text{cable},3} = 0$.

Step 3: Angular Accelerations → Gait Parameter Changes

Stride length change: The foot endpoint velocity determines stride length. Using the endpoint Jacobian:

$$\Delta\dot{\boldsymbol{p}}_{\text{toe}} = \boldsymbol{J}_{\text{toe}}(\boldsymbol{q}) \cdot \Delta\dot{\boldsymbol{q}} \quad \Rightarrow \quad \Delta\text{stride} \approx \int_{\text{swing}} \Delta\dot{x}_{\text{toe}}\,dt \tag{130}$$

Foot clearance change: The vertical position of the toe during swing determines ground clearance:

$$\Delta y_{\text{toe}} = \boldsymbol{J}_{\text{toe},y}(\boldsymbol{q}) \cdot \Delta\boldsymbol{q} = l_1 \sin\theta_1 \cdot \Delta\theta_1 + l_2 \sin\theta_2 \cdot \Delta\theta_2 + l_3 \sin\theta_3 \cdot \Delta\theta_3 \tag{131}$$

Step time change: The angular velocity of the thigh determines stride frequency:

$$\Delta T_{\text{step}} \approx -\frac{\Delta\dot{\theta}_1}{\dot{\theta}_1^2} \cdot \Delta\theta_{\text{stride}} \tag{132}$$

17.2 Complete Gait Cycle Analysis

The full EOM during cable-assisted walking:

$$\boxed{\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \underbrace{\boldsymbol{\tau}_{\text{muscle}}}_{\text{patient effort}} + \underbrace{\boldsymbol{J}_{\text{cable}}^T T_{\text{cable}}\hat{\boldsymbol{u}}}_{\text{H-Walker assist}} - \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{G}(\boldsymbol{q}) + \underbrace{\boldsymbol{J}_{\text{GRF}}^T \boldsymbol{F}_{\text{GRF}}}_{\text{stance only}}} \tag{133}$$

Swing Phase (60–100% GCP): Cable Assistance is Active

During swing, $\boldsymbol{F}_{\text{GRF}} = \boldsymbol{0}$:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = \boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{J}_{\text{cable}}^T T\hat{\boldsymbol{u}} \tag{134}$$

The cable force **reduces the muscle torque needed**:

$$\boldsymbol{\tau}_{\text{muscle}} = \underbrace{\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G}}_{\text{required for desired motion}} - \underbrace{\boldsymbol{J}_{\text{cable}}^T T\hat{\boldsymbol{u}}}_{\text{cable provides this}} \tag{135}$$

This is the **assist-as-needed** principle: the cable provides part of the required torque, reducing the patient's effort.

Stance Phase (0–60% GCP): Cable Maintains Pretension

During stance, the cable maintains minimum tension ($T = 5$ N) to avoid slack.

17.3 Control Strategies Using This Model

17.3.1 Strategy 1: Inverse Dynamics Feedforward (ILC)

Compute the required cable force to track a desired gait trajectory:

$$T^*_{\text{cable}}(t) = \frac{\left[\boldsymbol{M}\ddot{\boldsymbol{q}}_d + \boldsymbol{C}\dot{\boldsymbol{q}}_d + \boldsymbol{G} - \boldsymbol{\tau}_{\text{patient}}\right]_{\text{knee}}}{\left[\boldsymbol{J}^T_{\text{cable}}\hat{\boldsymbol{u}}\right]_{\text{knee}}} \tag{136}$$

Then ILC refines this across gait cycles:

$$T_{k+1}(t) = Q\left[T_k(t) + L \cdot e_k(t)\right] \tag{137}$$

where e_k is the tracking error in cycle k, L is the learning gain, and Q is a low-pass filter.

17.3.2 Strategy 2: Impedance Control with Model Compensation

Shape the apparent dynamics felt by the patient:

$$T_{\text{cable}} = \frac{1}{[\boldsymbol{J}^T_{\text{cable}}\hat{\boldsymbol{u}}]_{\text{knee}}}\Big[\underbrace{\boldsymbol{G}_2(\boldsymbol{q})}_{\text{gravity comp}} + \underbrace{K_v(\dot{\theta}_{2,\text{ref}} - \dot{\theta}_2)}_{\text{virtual damping}} + \underbrace{K_p(\theta_{2,\text{ref}} - \theta_2)}_{\text{virtual spring}}\Big] \tag{138}$$

The model provides $\boldsymbol{G}(\boldsymbol{q})$ for **gravity compensation** so the patient feels the leg is "weightless", requiring less effort to swing.

17.3.3 Strategy 3: MPC with Nonlinear Dynamics

Use the full model as the prediction model in MPC:

$$\min_{T_{\text{cable}}(t)} \quad \sum_{k=0}^{N} \left\|\boldsymbol{q}(t_k) - \boldsymbol{q}_{\text{ref}}(t_k)\right\|^2_{\boldsymbol{Q}} + \left\|T_{\text{cable}}(t_k)\right\|^2_R \tag{139}$$

$$\text{s.t.} \quad \boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = \boldsymbol{J}^T_{\text{cable}} T\hat{\boldsymbol{u}} \tag{140}$$

$$5 \le T_{\text{cable}} \le 70 \text{ N} \tag{141}$$

$$y_{\text{toe}}(\boldsymbol{q}) \ge y_{\text{clearance}} \text{ (foot clearance constraint)} \tag{142}$$

The model enables: trajectory prediction, constraint enforcement (max force, foot clearance), and optimal cable force profiling.

17.4 Sensitivity Analysis: How Force Direction Affects Gait

For a fixed attachment point, the cable direction $\hat{\boldsymbol{u}}$ changes with configuration. The **sensitivity** of each joint torque to cable tension is:

$$\boldsymbol{s}(\boldsymbol{q}) \triangleq \frac{\partial \boldsymbol{\tau}_{\text{cable}}}{\partial T} = \boldsymbol{J}^T_{\text{cable}}(\boldsymbol{q})\,\hat{\boldsymbol{u}}(\boldsymbol{q}) = \begin{bmatrix} s_1(\boldsymbol{q}) \\ s_2(\boldsymbol{q}) \\ 0 \end{bmatrix} \tag{143}$$

This vector answers: "*if I increase cable tension by 1 N at this instant, how much additional torque does each joint get?*"

The sensitivity depends on:

- **Cable angle** relative to each link: perpendicular force gives maximum moment arm, parallel gives zero.
- **Leg configuration**: as the leg swings, the cable angle changes, so the sensitivity changes throughout the gait cycle.

Practical Design Question

Given the fixed pulley position on the walker frame, at which GCP% is the cable most/least effective?
Answer: Compute $\|\boldsymbol{s}(\boldsymbol{q}(t))\|$ over one gait cycle. Peak sensitivity = best time to apply force. Minimum sensitivity = the cable geometry is unfavorable.
If the sensitivity is too low during a critical phase, the **pulley position should be redesigned**.

18 Summary

This document presented the complete nonlinear equations of motion for a three-link planar biomechanical model of the human lower limb. The key results are:

1. **Mass matrix** (eq. (34)): 3×3 symmetric positive definite, with constant diagonal and configuration-dependent off-diagonal elements.
2. **Coriolis matrix** (eq. (52)): Derived via Christoffel symbols to ensure the passivity property ($\dot{\boldsymbol{M}} - 2\boldsymbol{C}$ skew-symmetric).
3. **Gravity vector** (eqs. (57) to (59)): Three sinusoidal terms representing gravitational torques.
4. **Cable force mapping** (eq. (109)): Jacobian transpose mapping from cable tension to joint torques.
5. **Normal gait reference**: Fourier series with 6 harmonics for hip, knee, and ankle angles.

The model supports:

- Forward dynamics simulation (ODE integration)
- Inverse dynamics (feedforward torque computation)
- Energy analysis and conservation verification
- Cable-force-aware control design for the H-Walker
- Linearization for MPC and impedance control

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